

THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture – Introduction to Options

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Outline

- **Option types and positions**
- Moneyness, intrinsic and time value
- Factors affecting option prices
- Early exercise
- Upper and lower bounds
- Put-call parity

Reading: Chapters 9 & 10

Options

- A call option gives the holder the right to buy an asset by a certain date for a certain price (the strike price)
- A put option gives the holder the right to sell an asset by a certain date for a certain price (the strike price)

American vs. European options

- An American option can be exercised at any time during its life
- A European option can be exercised only at maturity

Options vs. futures/forwards

- In a futures/forward contract the holder has an obligation to buy or sell the asset at a certain price

- An option gives the holder the right to buy or sell the asset at a certain price

Notations

c: European call option price

p: European put option price

S_0: Stock price today

K: Strike price

T: Life of option

σ: Volatility of stock price

C: American Call option price

P: American Put option price

S_T : Stock price at option maturity

r: Risk-free rate for maturity T with continuous compounding

Option types and positions

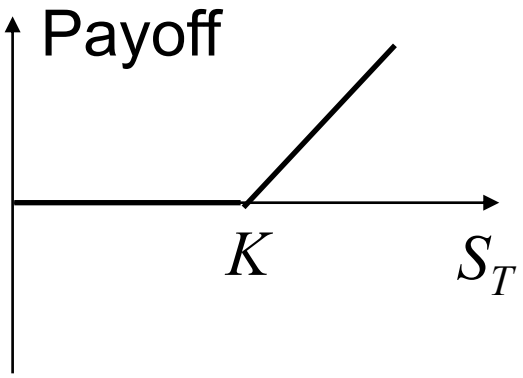
- Four positions
  - Long call
    - A long position in a call option, which means they are the holder of the call option and have the right but not the obligation to buy the asset. Their payoff is: $\max(S_T-K, 0)$. Their profit is: $\max(S_T-K, 0)-c$.
  - Short call
    - A short position in a call option, which means they are the writer of the call option and are obligated to sell the asset to the holder if the holder decides to exercise the option. Their payoff is: $\min(K- S_T, 0)$. Their profit is: $\min(K- S_T, 0)+c$

Option types and positions

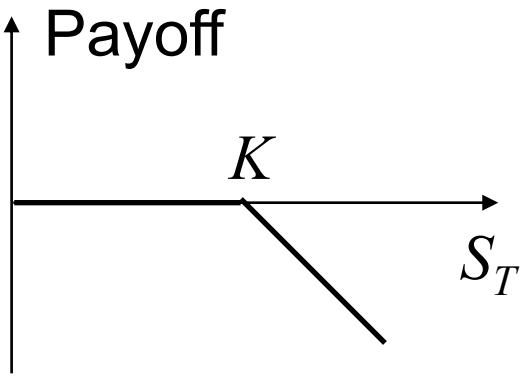
- Four positions
  - Long put
    - A long position in a put option, which means they are the holder of the put option and have the right but not the obligation to sell the asset. Their payoff is: $\max(K- S_T, 0)$. Their profit is: $\max(K- S_T, 0)-p$.
  - Short put
    - A short position in a put option, which means they are the writer of the put option and are obligated to buy the asset off the holder if the holder decides to exercise the option. Their payoff is: $\min(S_T-K, 0)$. Their profit is: $\min(S_T-K, 0)+p$.

Payoffs from Options

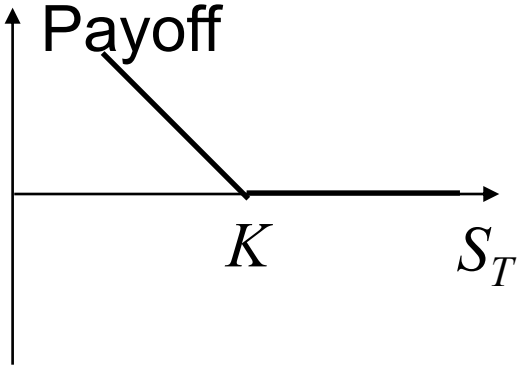
Long call



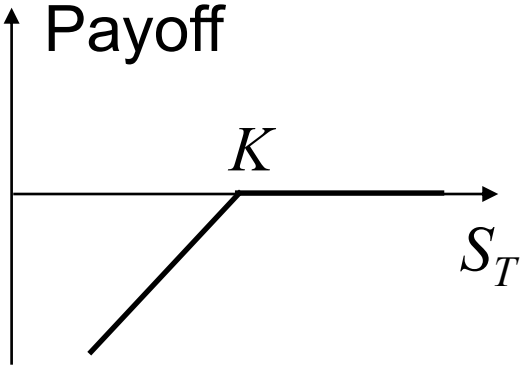
Short call



Long put

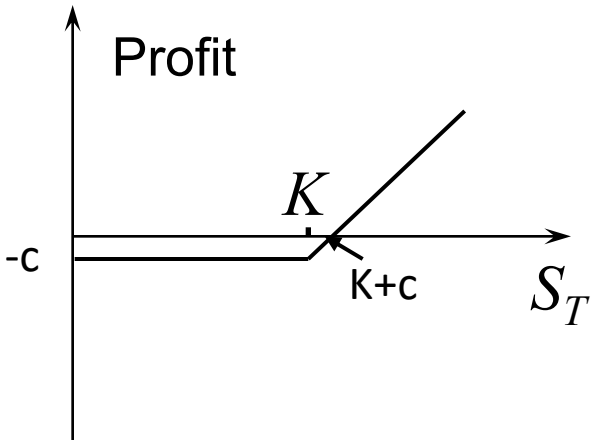


Short put

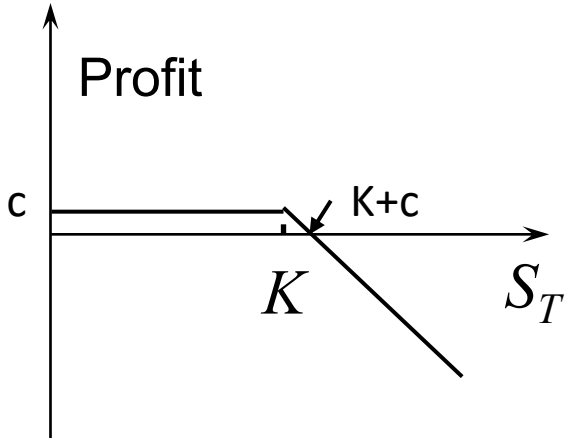


Profits from Options

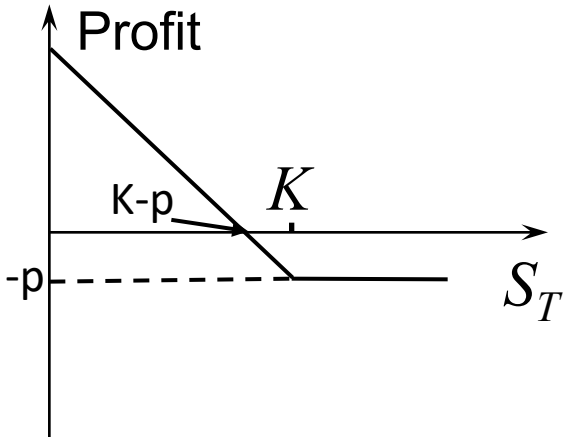
Long call



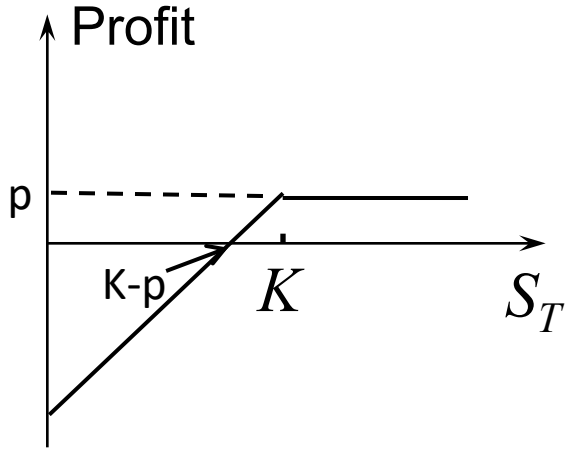
Short call



Long put



Short put



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Moneyness & Intrinsic Value

- Intrinsic value: max of zero & the value of option if exercised immediately
  - max(S – K, 0) for a call option
- Moneyness: the call option is
  - In the money if S > K
  - Out of the money if S < K
  - At the money if S = K

- Intrinsic value for put option is max(K - S, 0)
- Moneyness
  - In the money if S < K
  - Out of the money if S > K
  - At the money if S = K

Time value

- Option value = Intrinsic value + time value
- Time value:
  - captures possibility option may increase in value due to volatility in the underlying asset
  - positive, declines exponentially with time and reaches zero at maturity
  - Usually worthwhile for the holder of American option to wait rather than exercise immediately
    - if exercise receive intrinsic value & lose time value

https://www.asx.com.au/markets/company/BHP

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- Option types and positions
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- **Factors affecting option prices**
- Upper and lower bounds
- Put-call parity
- Early exercise
- The effect of dividends

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Factors Affecting Option Prices

- current stock price, S_0
- strike price, K
- time to expiration, T
- volatility of the underlying asset, σ
- risk-free rate, r
- dividends expected during the life of the option, D

	European		American	
Variable	c	p	C	P
S_0	+	-	+	-
K	-	+	-	+
T	?	?	+	+
σ	+	+	+	+
r	+	-	+	-
D	-	+	-	+

Factors Affecting Option Prices

Stock Price

- Calls more valuable as stock price **increases**
- Puts more valuable as stock price **decreases**

Strike Price

- Calls more valuable as strike price **decreases**
- Puts more valuable as strike price **increases**

Dividends

- Dividends reduce the stock price
- Value of call decreases and put increases

Risk free rate

- Increase in risk free rate decreases PV of K
- **Increases** value of call (pay less for the asset in todays $)
- **Decreases** value of put (receive less for the asset in todays $)

Volatility

- Higher vol increases chance stock will do very well or very poorly.
- You have no downside risk when you are long an option, so the chance of higher payoffs increases call and put value.

Factors Affecting Option Prices

Time to Expiration

- American: longer dated options have everything shorter dated options have plus more. As T increases put & calls more valuable.
- European call: Increase in T
  - Increases $\sigma\sqrt{T}$ increases value of call (greater dispersion of possible outcomes)
  - Decreases PV of K: increases value of call
  - If no dividend paid, increase value of call
  - If dividend paid, the decrease in spot: decrease value of call
  - Ambiguous if dividend paid
- European put: Increase in T
  - Increases $\sigma\sqrt{T}$ increases value of put (greater dispersion of possible outcomes)
  - Decreases PV of K: decreases value of put
  - If dividend paid, the decrease in spot: increase value of put
  - Ambiguous

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Never exercise an American call early (no dividends)

Assume XYZ pays **no dividends**. Share price is $70.

Consider a call option on XYZ with strike price of $40 with one month to expiration.

Given that the option is deep in the money, should a **hedger** exercise it immediately?

The answer is no.

- If you exercise the option and buy the stock now at $40 → Present value of the cost = $40.
- If you buy the stock one month later → Present value of the cost < $40.
- If stock price falls below $40 in one-month time → you will not exercise it and you should be glad that decision to exercise early was not taken. This reflects the time value of the option.

Never exercise an American call early (no dividends)

For a **speculator** who has no interest in owning the underlying asset, why not sell the option and realise a profit of $30? If we wait until T and S_T falls below $40, we will not exercise the option, so should we exercise it early?

The answer is no. To maximise the profit from trading the call option, the speculator should sell the option at t0 for a price greater than $30, because the option's value includes both its intrinsic value ($30) and time value.

May exercise an American call early (with dividends)

Assume XYZ pays dividends

- if the dividend amount is significant and exercising the option early to capture the dividend is economically advantageous, the option holder may choose to exercise the option before its expiration date

Early exercise of American put option

American put options may be exercised early when the underlying asset's price drops significantly and it becomes financially advantageous for the option holder to exercise the put option before its expiration date.

This applies to put options on dividend-paying and non-dividend-paying stocks.

Other possible reasons could be

- Stock price volatility decreases
- Interest rates increase

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Option Bounds

An American or European call option gives the holder the right to buy one share of a stock for a certain price. No matter what happens, the option can never be worth more than the stock. Hence, the stock price is an upper bound to the option price:

$$c \leq S_0 \text{ and } C \leq S_0$$

If this relationship did not hold an arbitrageur could easily make a riskless profit by buying the stock and selling the call option.

Option Bounds

An American or European put option gives the holder the right to sell one share of a stock for the exercise price. No matter how low the stock price becomes, the option can never be worth more than K. Hence:

$$p \leq K \text{ and } P \leq K$$

For European options, we know that at maturity the option cannot be worth more than K. It follows that it cannot be worth more than the present value of K today, as it cannot be exercised early. Hence:

$$p \leq Ke^{-rT}$$

Option Bounds

The lower bound for the price of a European call option on a non-dividend paying stock is:

$$c \geq \max(S_0 - Ke^{-rT}, 0)$$

We can prove this by considering the following two portfolios:

- Portfolio A
  - One European call option; plus,
  - An amount of cash equal to Ke^{-rT} or Ke^{-rT} invested in bond
- Portfolio B
  - One share

Option Bounds

- In portfolio A, if the cash is invested at the risk-free interest rate it will grow to K at time T.
  - If $S_T > K$, the call option is exercised at maturity and portfolio A is worth S_T.
  - If $S_T < K$, the call option expires worthless and the portfolio is worth K.
  - Hence at time T, portfolio A is worth max (S_T,K).

- Portfolio B is worth S_T at time T.
- Hence portfolio A is always worth at least as much as portfolio B at the options maturity. Thus this must also be true today. Also, as the worst that can happen to a call option is that it expires worthless, its value cannot be negative. Hence:

$$c \geq \max(S_0 - Ke^{-rT}, 0)$$

Option Bounds

The lower bound for the price of a European put option on a non-dividend paying stock is:

$$p \geq \max(Ke^{-rT}-S_0,0)$$

We can prove this by considering the following two portfolios:

- Portfolio C
  - One European put option; plus,
  - One share.
- Portfolio D
  - An amount of cash equal to Ke^{-rT}.

Option Bounds

- In portfolio C:
  - If $S_T<K$, the put option is exercised at maturity and portfolio C is worth K.
  - If $S_T>K$, the put option expires worthless and the portfolio is worth S_T.
  - Hence at time T, portfolio C is worth max (S_T,K).
- Assuming the cash in Portfolio D is invested at the risk-free interest rate, portfolio D is worth K at time T.
- Hence portfolio C is always worth at least as much as portfolio D at the options maturity. Thus this must also be true today. Also, as the worst that can happen to a put option is that it expires worthless, its value cannot be negative. Hence:
  - $p \geq \max(Ke^{-rT}-S_0,0)$

The Impact of Dividends on Lower Bounds for Options

$$c \geq \max(S_0 - D - Ke^{-rT}, 0)$$

$$p \geq \max(D + Ke^{-rT} - S_0, 0)$$

Option Bounds

Because American call options are never exercised early, they are equivalent to European call options, C=c.

$$C \geq \max(S_0 - Ke^{-rT}, 0)$$

American put options are sometimes exercised early.

$$P \geq \max(K - S_0, 0)$$

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Put-Call Parity (no dividend)

Consider the following two portfolios:

- Portfolio A: European call on a stock + PV of the strike price in cash
- Portfolio C: European put on the stock + the stock

Both are worth $\mathrm{MAX}(S_T, K)$ at the maturity of the options.

They must therefore be worth the same today. This means:

$$c + Ke^{-rT} = p + S_0$$

Extensions of Put-Call Parity

American options; $D = 0$,

$$S_0 - K < C - P < S_0 - Ke^{-rT}$$

European options; $D > 0$,

$$c + D + Ke^{-rT} = p + S_0$$

American options; $D > 0$,

$$S_0 - D - K < C - P < S_0 - Ke^{-rT}$$

You should know these formulas, and be able to use them in an exam, but you do not need to know how to derive them.

Next Week

Option Binomial Pricing Model